MAJOR PROJECT | PHASE 1

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AI Based Crop Recommendation

An Explainable Machine-Learning System for Data-Driven Crop Selection

TEAM MEMBERS

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UNDER THE GUIDANCE OF

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INTRODUCTION

Why Crop Selection Needs Intelligence

Agriculture remains the backbone of rural livelihoods, yet the choice of which crop to grow is still largely driven by tradition and instinct rather than the measurable condition of the soil and climate.

Soil nutrients (N, P, K), pH and weather jointly determine which crop will thrive on a given plot — machine learning can learn this mapping directly from real agronomic data and turn it into an actionable, explainable recommendation.

22 Crop types the model can recommend

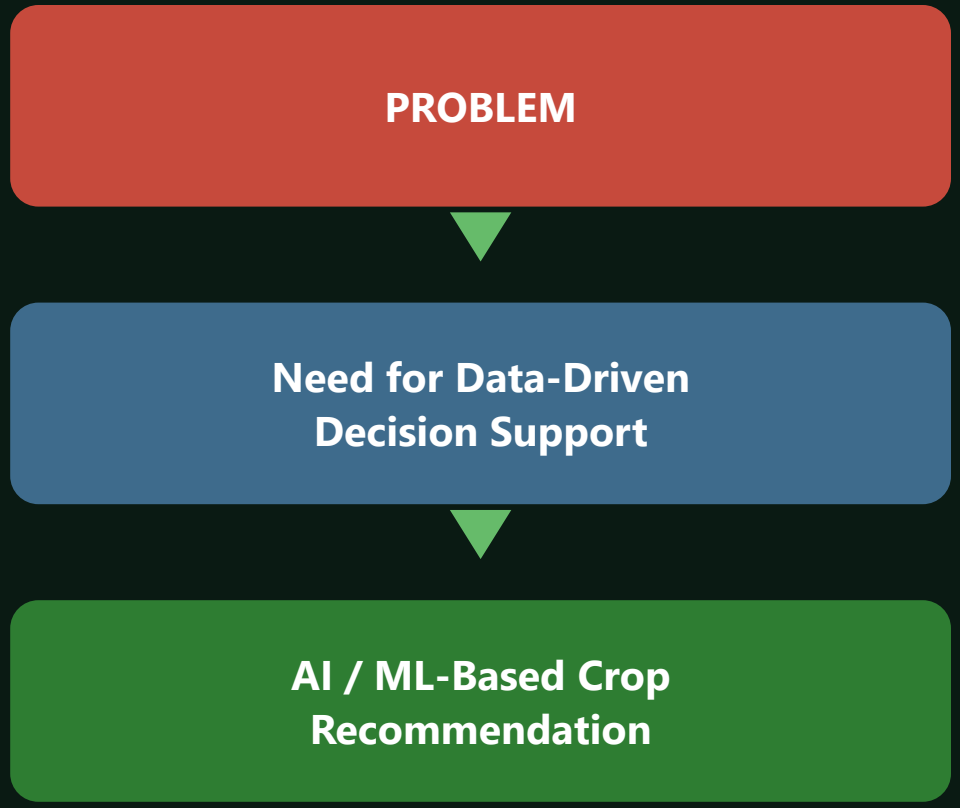
7 Soil + climate input parameters

99.3% Test accuracy of the trained model

PROBLEM STATEMENT

Farmers Lack Data-Driven Crop Guidance

- N — Limited visibility into soil nutrient levels (N, P, K) and pH
- ! — Crop choice relies on experience and guesswork, not measurement
- ~ — Environmental conditions (temperature, humidity, rainfall) keep changing
- x — Difficulty interpreting multiple agricultural parameters together
- $ — Wrong crop choice risks low productivity and financial loss



EXISTING APPROACH & LIMITATIONS

How Crop Selection Is Done Today

Approach	Limitation
Traditional / experience-based selection	Not repeatable, ignores measurable soil chemistry
Generic agronomy advisories	One-size-fits-all; not specific to a farmer's own soil report
Rule-based agri advisory tools	Hand-coded thresholds; brittle, don't generalize across regions
Black-box ML demos (where they exist)	Give a prediction with no explanation a farmer can trust

This motivates a system that learns directly from real soil-and-climate data, produces a ranked recommendation with a probability score, and explains why — the direction this project takes.

OBJECTIVES

What This Project Sets Out to Do

01

Build an ML-based crop recommendation system

Predict the most suitable crop from soil and environmental parameters (N, P, K, temperature, humidity, pH, rainfall).

02

Train and rigorously compare ML models

Evaluate Random Forest, Gradient Boosting, XGBoost and SVM with real accuracy, precision, recall and F1 — never a fabricated number.

03

Make recommendations explainable

Use SHAP feature-contribution analysis so a farmer sees not just the crop, but why it was recommended.

04

Provide data-driven decision support

Replace guesswork with a quantitative, reproducible recommendation pipeline.

05

Deliver a simple, farmer-oriented interface

Soil input form to ranked crop recommendation, in a few clicks.

06

Design for extensibility

Architect the system so future modules can be added without rewriting the core (see Future Scope).

LITERATURE REVIEW

20 Peer-Reviewed References Across 4 Themes

Minimum 5 verified papers per team member — organized by theme rather than presented as a flat list.

Ref. 1-5

Member 1

Crop Recommendation using Machine Learning

Ref. 6-10

Member 2

Foundational ML Algorithms Used in This Project

Ref. 11-15

Member 3

AI / Smart Agriculture Approaches

Ref. 16-20

Member 4

Explainable AI & Recent Intelligent Agriculture Systems

Crop Recommendation using Machine Learning

Ref. 1-5

#	Authors, Year & Title	Method	Key Result	Gap
1	Pudumalar et al. (2017) Crop Recommendation System for Precision Agriculture IEEE ICoAC 2016 (publ. 2017)	Ensemble voting (Random Tree, CHAID, KNN, NB)	Beats single classifiers	Region-specific, no explainability
2	Doshi et al. (2018) AgroConsultant: Intelligent Crop Recommendation System IEEE ICCUBEA 2018	RF, ANN, NB, SVM compared	RF/ensemble most accurate	Limited to Maharashtra data
3	Rajak et al. (2017) Crop Recommendation System to Maximize Crop Yield IRJET, Vol. 4	SVM, ANN, RF, Naive Bayes (comparative)	Multi-classifier crop recommendation	No feature explainability
4	Kulkarni et al. (2018) Improving Crop Productivity via Ensembling Technique IEEE CSITSS 2018	Ensemble: RF, Naive Bayes, Linear SVM	99.9% acc. on Kharif/Rabi crop data	Scale/generalizability untested
5	Bondre & Mahagaonkar (2019) Crop Yield & Fertilizer Recommendation using ML IJEAST, Vol. 4	Random Forest & SVM regression	RF outperformed SVM	Fertilizer logic is rule-based

Foundational ML Algorithms Used in This Project

Ref. 6-10

#	Authors, Year & Title	Method	Key Result	Gap
6	Breiman L. (2001) Random Forests Machine Learning, 45(1), 5-32	Bagged decision-tree ensemble	Lower variance, less overfitting	Basis of this project's active model
7	Friedman J.H. (2001) Greedy Function Approximation: A Gradient Boosting Machine Annals of Statistics, 29(5)	Stage-wise additive boosting	Defined the boosting framework used	Sensitive to hyperparameters
8	Chen T., Guestrin C. (2016) XGBoost: A Scalable Tree Boosting System ACM SIGKDD (KDD) 2016	Regularized gradient boosting	High accuracy/speed on tabular data	Needs careful tuning
9	Cortes C., Vapnik V. (1995) Support-Vector Networks Machine Learning, 20(3), 273-297	Max-margin kernel classification	Strong margin-based theory	Needs feature scaling, scales poorly
10	Kumar et al. (2015) Crop Selection Method to Maximize Crop Yield Rate IEEE ICSTM 2015	Crop Selection Method (CSM)	Better seasonal crop selection	No soil nutrient (N,P,K) features

AI / Smart Agriculture Approaches

Ref. 11-15

#	Authors, Year & Title	Method	Key Result	Gap
11	Liakos et al. (2018) Machine Learning in Agriculture: A Review Sensors (MDPI), 18(8):2674	Systematic ML-in-agriculture review	ML beats traditional statistics	Explainability, usability gaps noted
12	Kamilaris A., Prenafeta-Boldu F.X. (2018) Deep Learning in Agriculture: A Survey Computers & Electronics in Agri., 147	Survey of deep-learning applications	Improves accuracy, needs big data	Data scarcity, low interpretability
13	Suresh et al. (2021) Efficient Crop Yield Recommendation for Digital Farming Int'l J. of Modern Agriculture, 10(1)	SVM-based crop recommendation	Higher precision on soil+yield data	Single-region dataset only
14	Reddy D.J., Kumar M.R. (2021) Crop Yield Prediction Using Machine Learning Algorithm IEEE ICICCS 2021	RF vs. SVM vs. ANN comparison	RF gave best yield accuracy	No explainability layer
15	Sharma A., Jain A., Gupta P., Chowdary V. (2021) ML Applications for Precision Agriculture: A Review IEEE Access, Vol. 9, 4843-4873	Comprehensive ML-in-agri review	Maps ML methods to farm tasks	Few real-world deployments studied

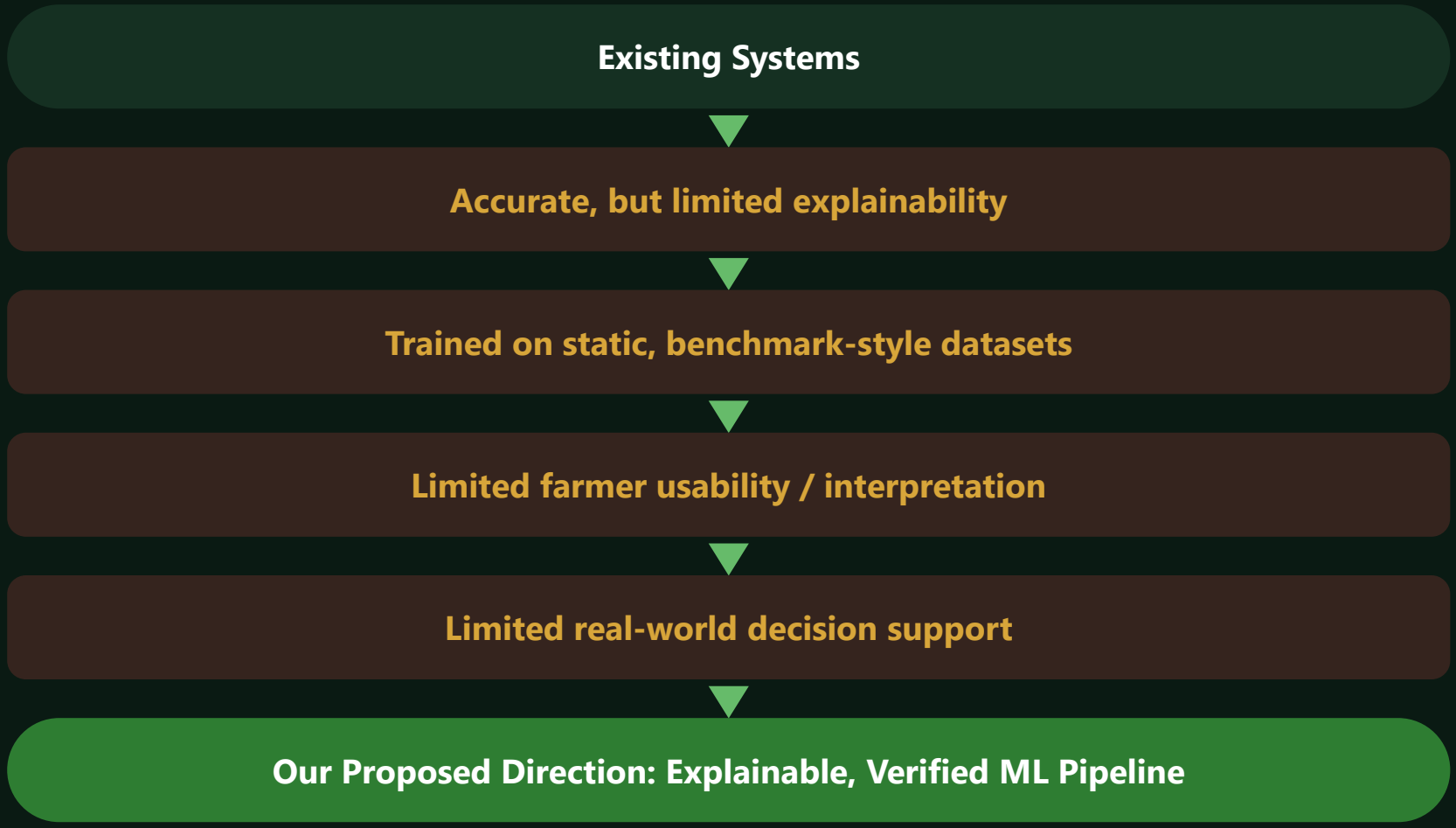
Explainable AI & Recent Intelligent Agriculture Systems

Ref. 16-20

#	Authors, Year & Title	Method	Key Result	Gap
16	Ribeiro M.T., Singh S., Guestrin C. (2016) "Why Should I Trust You?" Explaining Predictions of Any Classifier ACM SIGKDD (KDD) 2016	LIME: local surrogate explanations	Model-agnostic, increases trust	Local approximation, not global
17	Lundberg S.M., Lee S.-I. (2017) A Unified Approach to Interpreting Model Predictions NeurIPS 2017, Vol. 30	SHAP: Shapley-value attribution	Unifies prior explainability methods	Used directly in this project (SHAP)
18	Jeong J.H. et al. (2016) Random Forests for Global & Regional Crop Yield Predictions PLOS ONE, 11(6):e0156571	Random Forest yield regression	Beats traditional regression	Limited regional transferability
19	Patel K., Patel H.B. (2020) A State-of-the-Art Survey on Recommendation System and Prospective Extensions Computers and Electronics in Agriculture, Vol. 178	Survey of recommendation systems (incl. agriculture)	Maps techniques to farm use-cases	Calls for explainable, personalized RS
20	Priya P., Muthaiah U., Balamurugan M. (2018) Predicting Yield of the Crop Using Machine Learning IJESRT, Vol. 7(1)	Random Forest regression	Beats linear regression baseline	No interactive farmer-facing system

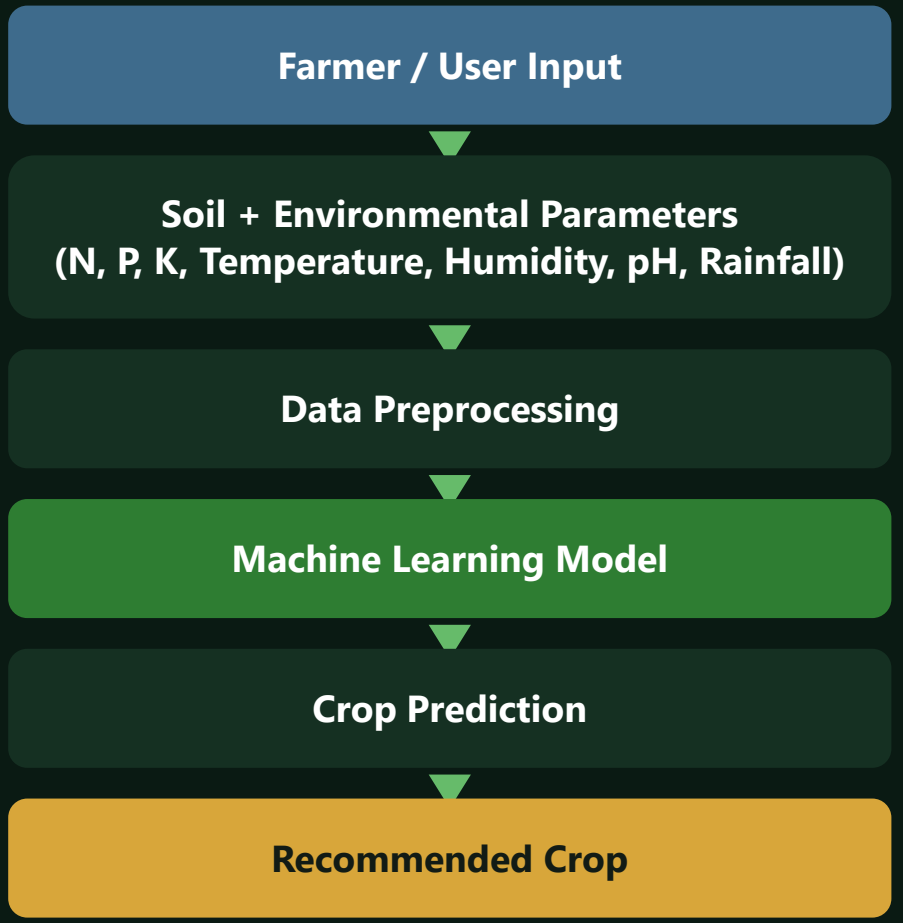
RESEARCH GAP

What the Literature Consistently Misses



PROPOSED SYSTEM

Data-Driven Crop Recommendation Workflow



DATASET

Crop Recommendation Dataset

Dataset name	Crop Recommendation Dataset
Source	Kaggle: atharvaingle/crop-recommendation-dataset (local CSV mirror)
Records	2,200 rows, 0 missing, 0 duplicates
Crop classes	22 crops, 100 samples each
Input features	7 numeric: N, P, K, temp, humidity, pH, rainfall
Target / label	Crop name (22 classes)
Data type	Structured / tabular

N	P	K	temp	humidity	ph	rainfall	label
90	42	43	20.9	82.0	6.50	202.9	rice
85	58	41	21.8	80.3	7.04	226.7	rice
71	54	16	22.6	63.7	5.75	87.8	maize
20	67	20	17.0	16.9	7.63	88.6	chickpea

Sample rows from ml/datasets/crop_recommendation.csv

Why this dataset is suitable

- Matches farmer-supplied soil + weather inputs directly
- Balanced: 100 samples/crop, no imbalance handling needed
- Retrains fast; large enough for a valid 80/20 split
- Widely used benchmark — enables literature comparison

DATASET

Feature Ranges & Crop Classes

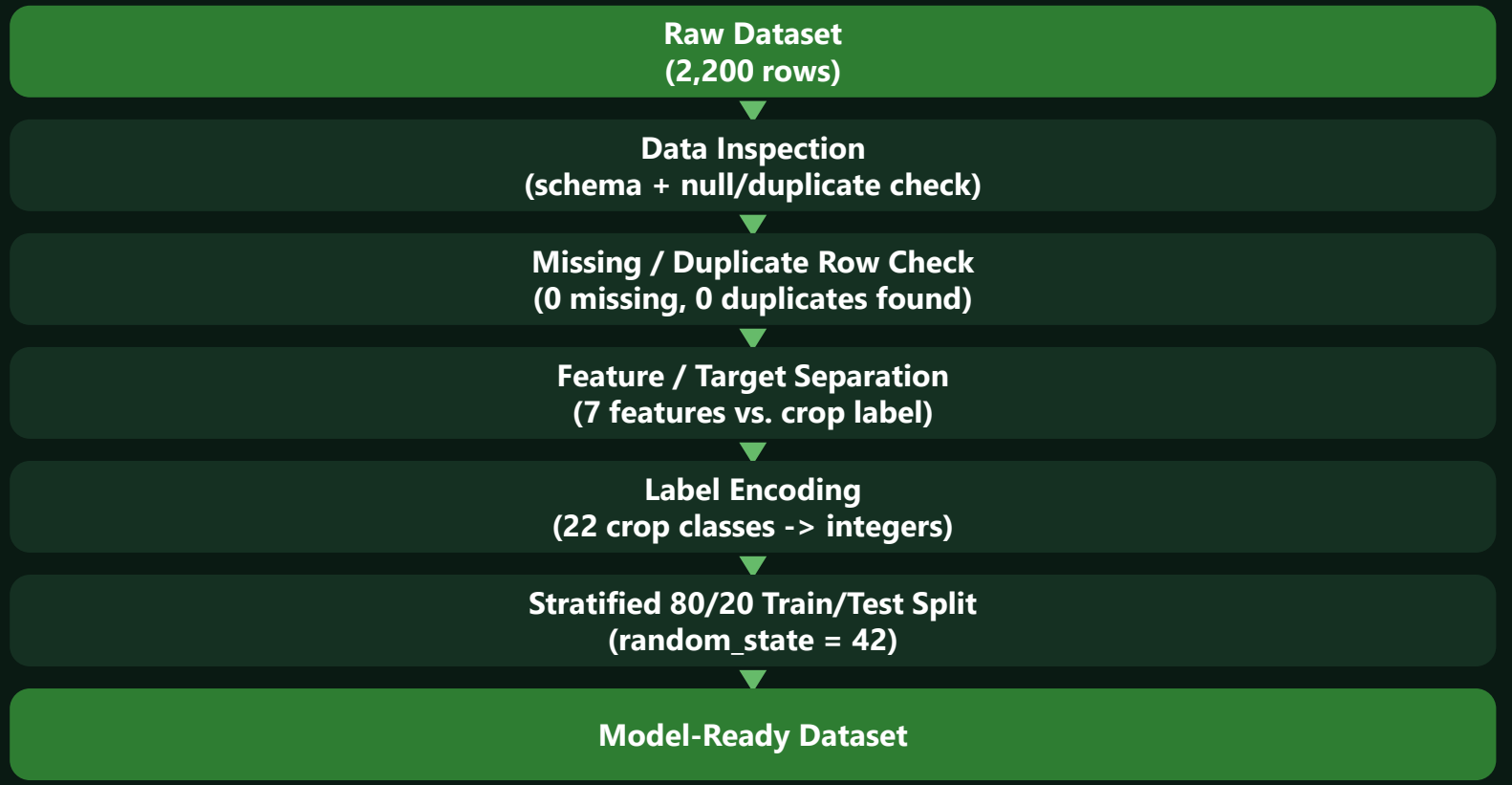
Feature	Meaning	Typical Range
N	Nitrogen content in soil (kg/ha)	0 - 140
P	Phosphorus content in soil (kg/ha)	5 - 145
K	Potassium content in soil (kg/ha)	5 - 205
temperature	Ambient temperature (deg C)	9 - 44
humidity	Relative humidity (%)	14 - 100
ph	Soil pH	3.5 - 10
rainfall	Rainfall (mm)	20 - 300

22 Crop Classes

rice	maize	chickpea
kidneybeans	pigeonpeas	mothbeans
mungbean	blackgram	lentil
pomegranate	banana	mango
grapes	watermelon	muskmelon
apple	orange	papaya
coconut	cotton	jute
coffee		

DATA PREPROCESSING

From Raw CSV to Model-Ready Data



No scaling / normalization / SMOTE / PCA is applied — Random Forest, Gradient Boosting and XGBoost are scale-invariant. SVM is trained on raw features in this MVP (documented limitation).

ML MODELS

Random Forest — The Active Model

Random Forest is an ensemble of many decision trees, each trained on a random subset of rows and features. The final prediction is the majority vote across all trees.

- **Why suitable here** — Handles non-linear soil/climate interactions; robust to noisy, mixed-scale tabular features like N,P,K vs. rainfall.
- **How it predicts** — 300 trees vote independently on the crop class; the class with the most votes (highest average probability) wins.
- **Advantages** — High accuracy, resistant to overfitting, gives feature importance for explainability, no scaling required.
- **Limitations** — Larger model size than a single tree; less interpretable per-tree (addressed here using SHAP).

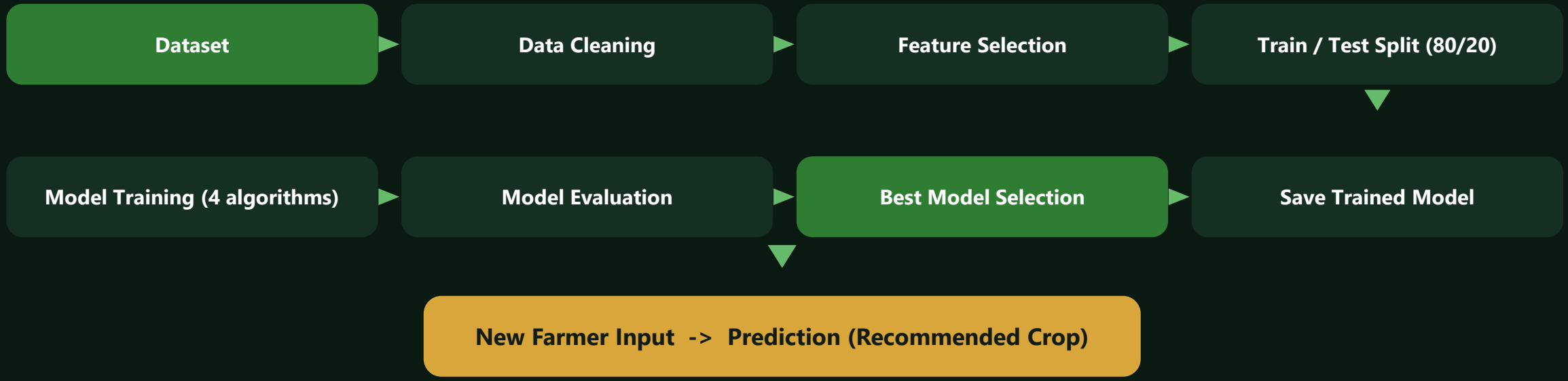
Model Comparison (Experimentation)

Random Forest	99.32%
Gradient Boosting	98.86%
XGBoost	98.64%
SVM	98.41%

All 4 trained & saved for transparent comparison. Random Forest is the active model used for live prediction (highest test accuracy).

TRAINING & TESTING

Pipeline: Dataset to Prediction



Training

Learns the relationship between soil/climate features and crop labels from the training split.

Testing

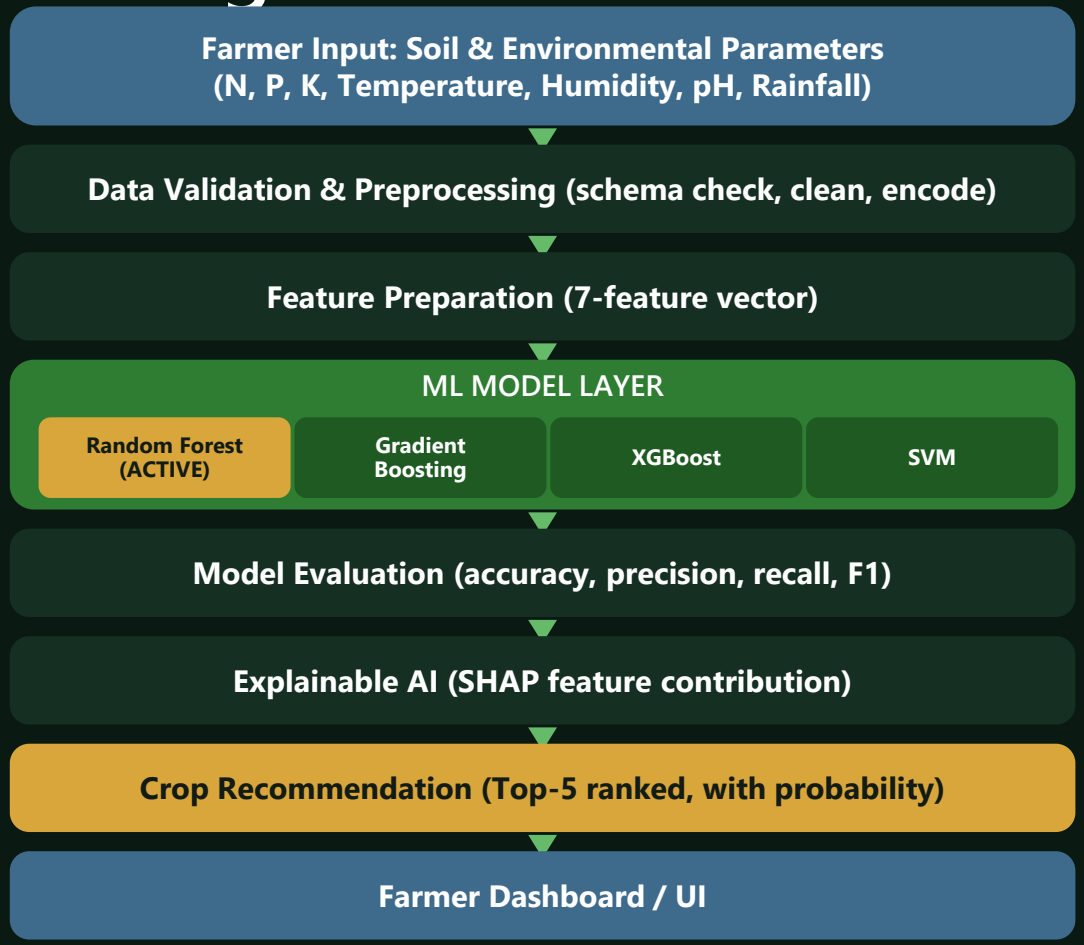
Evaluates the trained model on the unseen 20% split to measure real generalization.

Prediction

The saved model receives new soil/environmental values and returns a ranked crop recommendation.

SYSTEM ARCHITECTURE

End-to-End Block Diagram

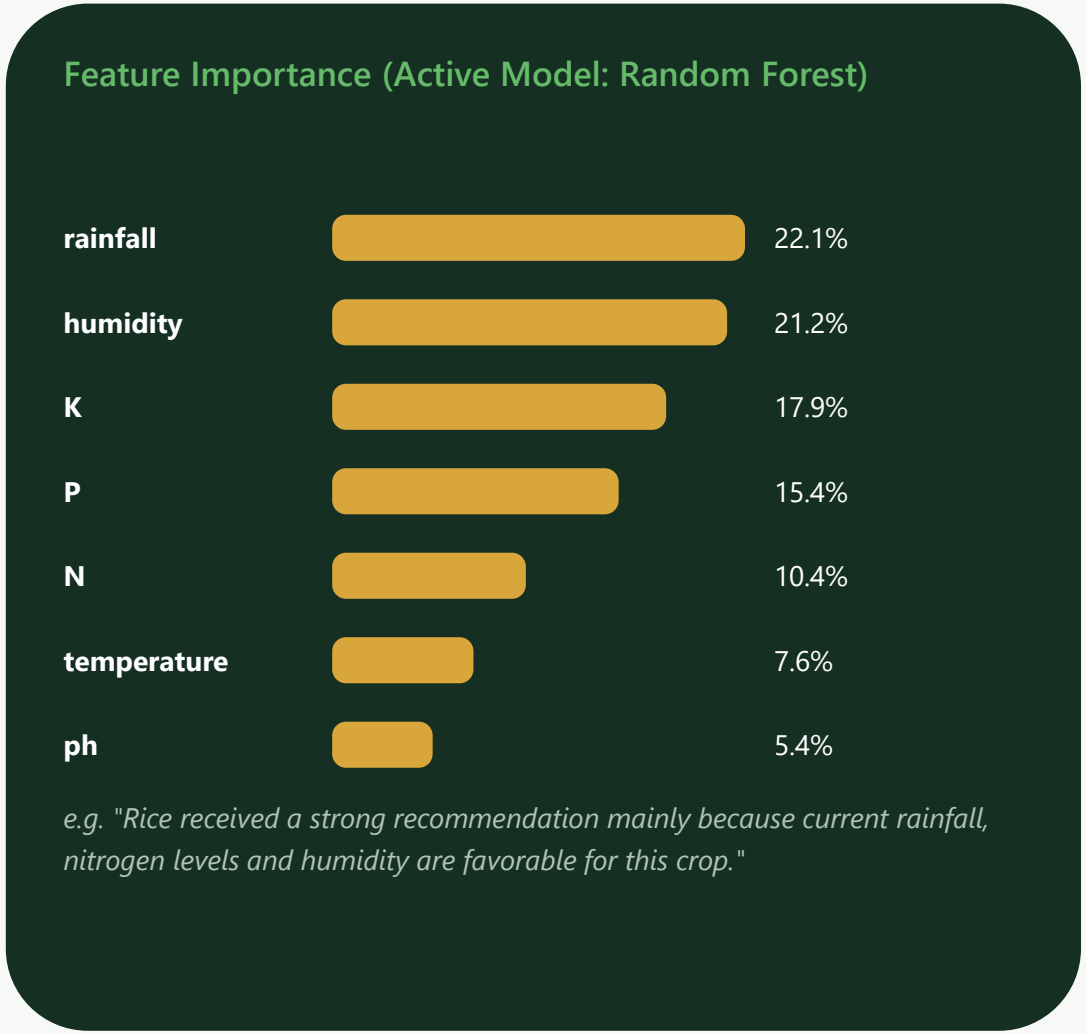


EXPLAINABILITY

Why This Crop? — SHAP-Based Explanation

Every recommendation is explained using SHAP (SHapley Additive exPlanations) applied to the active tree-based model — not a templated or hand-written justification.

1. Build the 7-feature input vector for the requested crop
2. Run SHAP TreeExplainer on the trained model
3. Get a real per-feature contribution value
4. Sort contributions by magnitude (largest influence first)
5. Generate a natural-language explanation from the top features



CURRENT IMPLEMENTATION — PHASE 1

What Is Working Today (Phase-1 Baseline)

- ✓ JWT-based authentication (register / login)
- ✓ Farm creation & soil parameter input form
- ✓ Farmer dashboard with soil health snapshot
- ✓ Crop recommendation engine (trained Random Forest, real inference)
- ✓ Top-5 ranked crops with real model probabilities
- ✓ Explainable AI: SHAP feature-contribution chart + generated explanation
- ✓ Model comparison lab: 4 trained algorithms, real metrics, confusion matrix
- ✓ Recommendation history log per farm

Extension Modules Delivered

9 / 10

Baseline crop recommendation system verified end-to-end. 9 of the 10 originally-planned extension modules (see next slide) are now implemented and tested — only Satellite/NDVI remains future scope.

Under Development / Not Yet Wired to ML

- Live weather data integration (currently mock/demo provider)
- Region/soil-type-specific model retraining
- Deployment-grade UI polish & user testing
- Satellite/NDVI monitoring (no data provider connected)

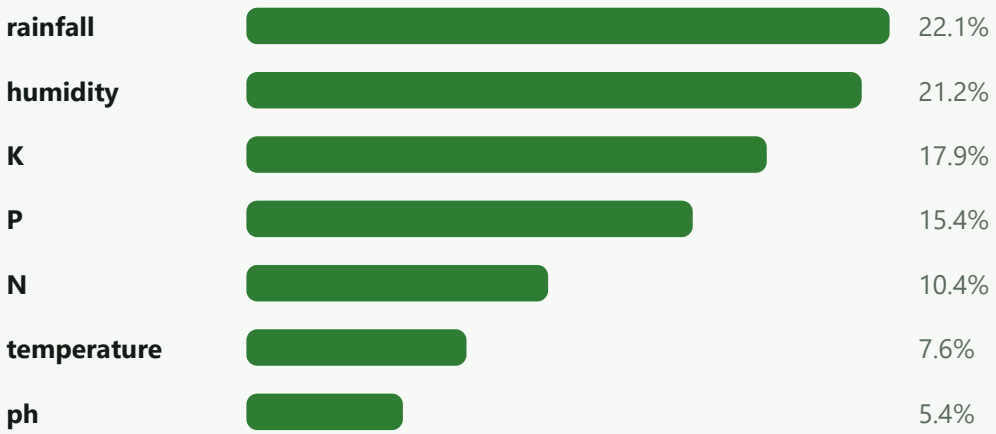
RESULTS & EVALUATION

Model Performance on Held-Out Test Data

Stratified 80/20 split (1,760 train / 440 test rows), random_state=42 — metrics from models/metrics.json

Model	Accuracy	Precision	Recall	F1-Score	5-Fold CV Mean
Random Forest (active)	99.32%	99.35%	99.32%	99.32%	99.41% (+/-0.27%)
Gradient Boosting	98.86%	98.97%	98.86%	98.87%	99.09% (+/-0.32%)
XGBoost	98.64%	98.74%	98.64%	98.62%	99.32% (+/-0.20%)
SVM (RBF)	98.41%	98.66%	98.41%	98.39%	97.82% (+/-0.40%)

Random Forest Feature Importance



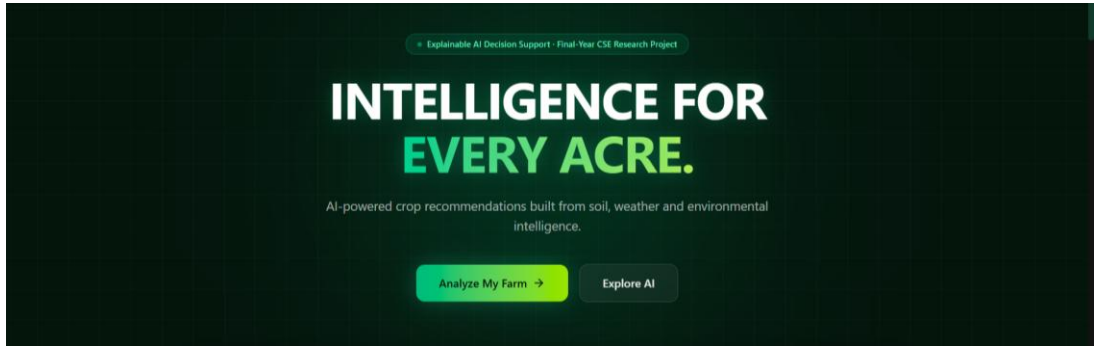
Honest Reading of These Numbers

This benchmark dataset is clean and perfectly balanced (100 samples/class), which is why all 4 models score 98-99%. Real field data is noisier — validating on messier, region-specific data is planned as the model matures beyond Phase 1.

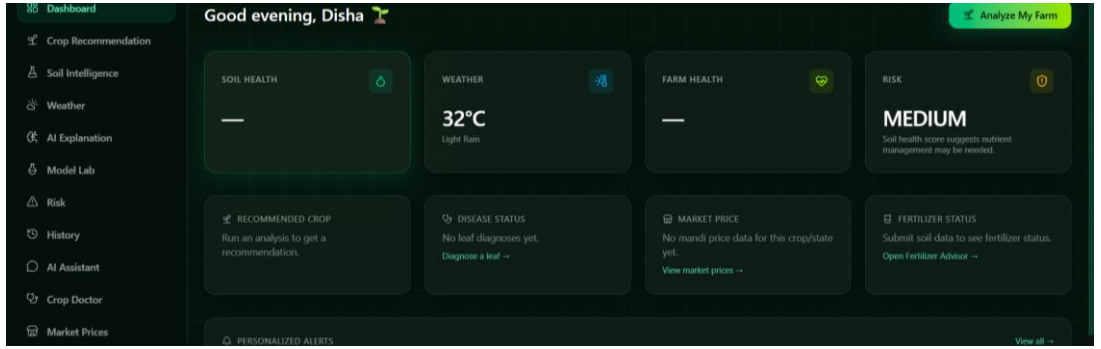
RESULT SCREENSHOTS

Application Walkthrough (Live Screens)

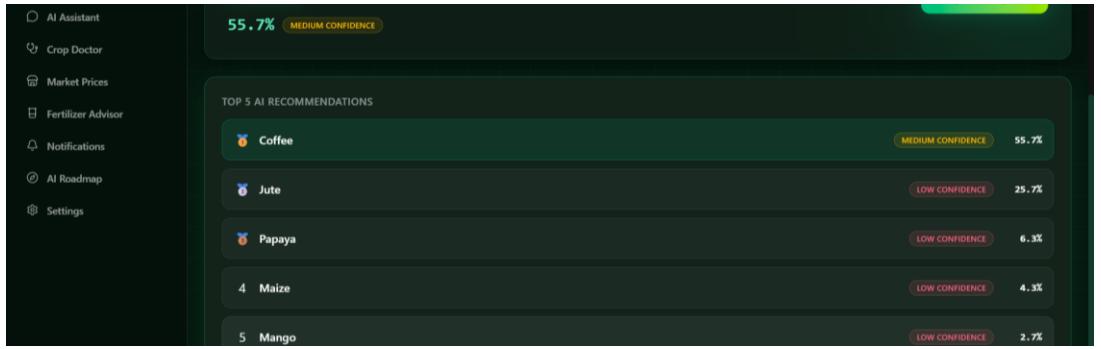
Introduction — System Landing Page



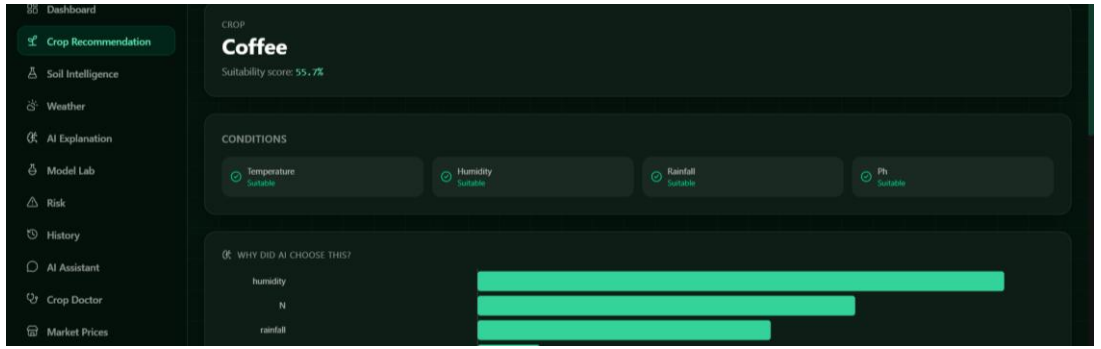
Farmer Dashboard



Crop Recommendation — Top-5 AI Results



Explainable AI — "Why This Crop?"



LIMITATIONS

Honest Constraints of the Current Build

Benchmark dataset	Clean, perfectly balanced (100 samples/crop); real field data is noisier, which inflates today's accuracy numbers.
No soil-type / location signal	Soil type, location and irrigation availability are collected in the UI but not yet used as model inputs.
SVM not feature-scaled	SVM is trained on raw features in this MVP; a StandardScaler step is the documented next improvement.
Single global model	One model for all regions — not yet specialized to India's diverse agro-climatic zones.
Weather is farmer-supplied	Temperature / humidity / rainfall are entered values in this phase, not yet a live weather feed.

IMPLEMENTATION STATUS UPDATE

Beyond Phase 1

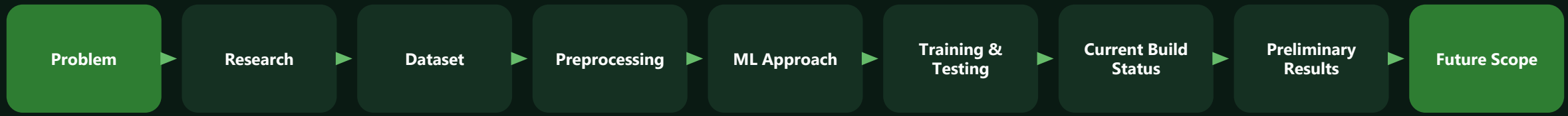
The architecture was designed to be extensible — 9 of these 10 modules are now implemented and verified end-to-end (real trained ML models, published formulas, or rule-based logic on real farm data — never fabricated). Only Satellite/NDVI has no data provider connected and remains future scope; physical IoT hardware is also not connected (the real ingestion endpoint works with demo sensor data).

✓ Leaf Disease Detection	✓ Crop Market Price Intelligence	✓ Fertilizer Recommendation	✓ Personalized Farmer Messages	✓ Yield Prediction
✓ Water Requirement Estimation	✓ Risk Prediction	✓ Profit Intelligence	✓ AI Agriculture Assistant	Real-Time Agri Intelligence (Weather / Satellite / IoT)

9 OF 10 MODULES IMPLEMENTED

CONCLUSION

Summary & Path Forward



AgriAI's crop recommendation baseline is real, trained and verified end-to-end: a public benchmark dataset, four honestly-compared ML models, a 99.3% test-accuracy Random Forest as the active model, and SHAP-based explainability. Since then, 9 of the 10 originally-planned extension modules — yield, profit, water, risk, disease detection, fertilizer, market price, notifications and the AI assistant — have also been implemented and verified end-to-end. Only satellite/NDVI monitoring remains future scope.

References

Full reference list: 20 literature-review papers (slides 7-10) + project documentation (docs/dataset.md, docs/ml-methodology.md, docs/architecture.md, docs/xai.md, docs/future-scope.md) + ml/crop_recommendation/train.py + models/metrics.json.